Credit Card Fraud Detection

Applied Machine Learning for Predictive Analytics

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Contents / Agenda

- Requirements Gathering
- Data Overview
- Exploratory Data Analysis
- Anomaly Detection
- Fraud Detection - Model Building
- Fraud Detection - Model Performance Comparison and Final Model Selection
- Business Insights and Recommendations

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Requirements Gathering

Business problem:

- The organization's current transaction authorization system is struggling to keep pace with increasingly sophisticated fraudulent behaviors that manifest as subtle deviations from typical customer patterns. High-volume, real-time activities—such as rapid transaction bursts, sudden geographic shifts, and the use of unrecognized devices—are often indistinguishable from legitimate behavior when using traditional, static rule-based systems. These rigid controls fail to capture the dynamic nature of modern fraud, leaving the digital payment platform vulnerable to financial losses and evolving criminal tactics across its international operations.
- A high volume of false declines, directly harms the customer experience. Legitimate transactions incorrectly blocked, causes customer friction and forces fraud analysts to manually intervene in low-risk scenarios. The business requires a more sophisticated approach that balances effective fraud loss prevention and regulatory compliance, ensuring that security measures do not come at the expense of customer satisfaction.

Objective:

Design and implement a scalable, behavior-driven fraud detection framework that utilizes anomaly detection and machine learning to distinguish between fraudulent patterns and legitimate customer activity in real time. This solution aims to minimize false positives and reduce fraud losses by replacing static rules with interpretable models that provide risk-based decisioning, such as automated approvals, step-up verifications, or immediate blocks.

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Requirements

1. How do customer behaviour and transaction context influence the likelihood of fraudulent activity?

-Customer behavior and transaction context significantly influence the likelihood of fraudulent activity by providing subtle behavioral deviations from normal patterns. Key behavioral factors include rapid transaction bursts, sudden changes in geographic location (impossible travel), and usage from new, unrecognized devices.

-The transaction context, such as the merchant category (e.g., elevated-risk merchants), the channel used (in-store, online, or cross-border), and the real-time location, also increases or decreases the perceived risk. By incorporating these specific behavioral and contextual signals, the fraud detection solution aims to improve identification accuracy and minimize false declines.

2. How do key transactional and risk-related indicators affect fraud detection accuracy and the ability to distinguish legitimate from suspicious events?

- Key transactional and risk-related indicators affect fraud detection accuracy by providing crucial signals that help distinguish between legitimate and suspicious events. Specific indicators like rapid transaction bursts, sudden location changes, and usage from new devices highlight subtle behavioral deviations that machine learning models use to identify anomalies effectively, especially those missed by static rules alone.

-By incorporating these detailed indicators, the system can better balance fraud loss prevention with customer experience and compliance. This approach prevents overly strict controls that lead to costly false declines. Ultimately, the accurate use of these indicators ensures that the fraud detection solution is effective at real-time identification while maintaining operational feasibility and explainability for risk-based decision-making.

3. Once a base fraud detection model is built, how do we assess and enhance its ability to identify fraudulent transactions in real time?

- To assess and enhance a real-time fraud detection model, organizations must implement low-latency performance monitoring and shadow mode testing. Monitoring focuses on metrics beyond standard accuracy, such as Precision-Recall curves and the False Positive Ratio (FPR), to ensure high-risk transactions are flagged without disrupting legitimate customer journeys.

4. How do hyperparameter tuning strategies impact the accuracy, responsiveness, and robustness of fraud prediction models?

- Organizations can precisely calibrate model parameters—such as tree depth and regularization—to maximize the detection of subtle fraud patterns while minimizing false declines. These techniques are particularly effective at handling imbalanced datasets, ensuring the model remains sensitive to rare fraudulent "bursts" without over-fitting to historical data. This leads to higher F1-scores and more reliable performance across diverse merchant categories and geographic locations.

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Requirements

5. What transactional patterns does the ML model rely on most to differentiate fraudulent from legitimate behavior?

- Temporal and Velocity Patterns: Models monitor the frequency, speed, and sequence of transactions. Key indicators include rapid transaction bursts (e.g., five transactions in 15 minutes), multiple small "test" purchases followed by a large one, or a sudden spike in activity that deviates from a customer's personalized baseline.

-Geographic and Device Intelligence: Models rely on device fingerprinting and location data to spot "impossible travel" scenarios—such as a card being used in two far-apart locations within an unfeasible timeframe.

6. Which transaction behaviors and anomaly scores trigger different real-time actions, and how do fraud risk and abnormal behavior combine to guide decisions?

- Organizations typically use a layered threshold system to manage transactions:

Automatic Approval (Low Risk/Normal Behavior): Transactions with low anomaly and risk scores (e.g., <15 on a 0-100 scale) proceed without friction.

Step-Up Verification / MFA (Medium Risk/Abnormal Behavior): Triggered by behavioral deviations such as a sudden large purchase at an unusual time (e.g., 3 a.m.) or from a new device. In 2026, this increasingly involves biometric face scans or device confirmations.

Manual Review (High Risk/Suspicious Patterns): Moderately high scores (e.g., 10-20 or higher depending on the platform) may pause the user journey for an analyst's investigation if the transaction doesn't meet strict blocking criteria.

Immediate Block (Very High Risk/Known Fraud Signs): Scores exceeding a critical threshold (e.g., >85 or 95) result in an instant decline. This is often triggered by impossible travel (geographic distance inconsistencies) or repeated velocity-based failures.

Example decision framework

Anomaly Score	Fraud Probability	Action
Low	Low	Automatic Approve
High	Low / Medium	Verify
Low / Medium	High	Manual Review
High	High	Block transaction

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Data Overview

- **Shape of the dataset.**
  The dataset has 60000 rows and 32 columns.
- **Attribute data types of the dataset.**
  The dataset has the following data types:
  float64(6), int64(18), object(8)
- **Duplicate values in the dataset.**
  There are no duplicate values in the data
- **Missing values in the dataset**
  There are no missing values in the dataset.

Statistical summary of the dataset.

Transaction Volume and Scale: The dataset contains 60,000 transactions across 924 unique cards and 3,000 merchants. Transaction amounts average $198.63 but show significant variance with a maximum value of $3,833.26, suggesting a mix of routine retail and high-value purchases.

Dominant Channels and Methods: Point of Sale (POS) is the primary transaction channel (35,385 occurrences), and Chip is the leading entry mode (28,241 occurrences). Security checks appear robust, as CVV matches occurred in over 97% of transactions (58,647).

High-Risk Geographic and Device Signals: While average distances between transactions are low (67 km), the maximum distance is 2,999 km, highlighting potential "impossible travel" scenarios. Additionally, 2% of transactions involve a new device, and 1% show a country mismatch between the device and the transaction.

Velocity and Temporal Patterns: Short-term transaction frequency is generally low (averaging less than 1 per minute), but some accounts show rapid bursts of up to 4 transactions in a single minute or 6 in five minutes, which are key indicators for behavioral fraud detection.

Merchant Risk Profile: The average merchant fraud rate is approximately 3%, but some merchants have fraud rates as high as 30%. Furthermore, 1.6% of transactions are with new merchants, and 2.8% are cross-border, both of which represent elevated risk categories.

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Statistical summary of the dataset.

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
transaction_id	60000.0	NaN	NaN	NaN	30000.5	17320.652413	1.0	15000.75	30000.5	45000.25	60000.0
account_id	60000.0	NaN	NaN	NaN	100224.124667	130.485308	100000.0	100109.0	100227.0	100336.0	100448.0
card_id	60000	924	card_100265_0	94	NaN	NaN	NaN	NaN	NaN	NaN	NaN
txn_hour	60000.0	NaN	NaN	NaN	11.899617	6.643373	0.0	6.0	12.0	17.0	23.0
txn_day_of_week	60000.0	NaN	NaN	NaN	3.102233	2.998171	0.0	0.0	6.0	6.0	6.0
transaction_amount	60000.0	NaN	NaN	NaN	198.630492	225.909078	1.0	76.639876	150.22	247.153594	3833.256402
channel	60000	3	POS	35385	NaN	NaN	NaN	NaN	NaN	NaN	NaN
entry_mode	60000	4	chip	28241	NaN	NaN	NaN	NaN	NaN	NaN	NaN
txn_count_1min	60000.0	NaN	NaN	NaN	0.619083	0.751107	0.0	0.0	1.0	1.0	4.0
txn_count_5min	60000.0	NaN	NaN	NaN	1.563433	1.053484	0.0	1.0	2.0	2.0	6.0
avg_txn_amount_30d	60000.0	NaN	NaN	NaN	149.978688	84.553009	10.01	76.71	144.73	221.31	299.6
max_txn_amount_90d	60000.0	NaN	NaN	NaN	824.559805	535.729437	37.85	386.07	745.25	1206.82	2348.18
merchant_id	60000	3000	m_1576	40	NaN	NaN	NaN	NaN	NaN	NaN	NaN
merchant_category_code	60000.0	NaN	NaN	NaN	5653.502383	256.251126	5311.0	5411.0	5732.0	5812.0	5999.0
merchant_fraud_rate_30d	60000.0	NaN	NaN	NaN	0.030174	0.034527	0.0	0.013	0.026	0.039	0.3
is_new_merchant	60000.0	NaN	NaN	NaN	0.016867	0.128773	0.0	0.0	0.0	0.0	1.0
merchant_country	60000	5	FR	12381	NaN	NaN	NaN	NaN	NaN	NaN	NaN
cardholder_country	60000	5	FR	12382	NaN	NaN	NaN	NaN	NaN	NaN	NaN
cross_border_flag	60000.0	NaN	NaN	NaN	0.028233	0.16564	0.0	0.0	0.0	0.0	1.0
distance_from_last_txn_km	60000.0	NaN	NaN	NaN	67.085921	271.152467	0.0	12.72	25.65	38.42	2999.07
time_since_last_txn_sec	60000.0	NaN	NaN	NaN	1770.75215	1057.339094	30.0	842.0	1775.0	2689.0	3599.0
device_id	60000	1593	dev_883	179	NaN	NaN	NaN	NaN	NaN	NaN	NaN
is_new_device	60000.0	NaN	NaN	NaN	0.020417	0.141422	0.0	0.0	0.0	0.0	1.0
device_country_mismatch	60000.0	NaN	NaN	NaN	0.009967	0.099335	0.0	0.0	0.0	0.0	1.0
cvv_result	60000	2	match	58647	NaN	NaN	NaN	NaN	NaN	NaN	NaN
cvv_fail_count_24h	60000.0	NaN	NaN	NaN	0.061433	0.443166	0.0	0.0	0.0	0.0	5.0
card_age_days	60000.0	NaN	NaN	NaN	1037.996817	863.220453	30.0	304.0	828.0	1579.0	3902.0
account_age_days	60000.0	NaN	NaN	NaN	2058.929083	1127.460535	104.0	1120.0	2141.0	3037.0	3997.0
historical_fraud_flag	60000.0	NaN	NaN	NaN	0.0995	0.299335	0.0	0.0	0.0	0.0	1.0
anomaly_label	60000.0	NaN	NaN	NaN	0.1	0.300003	0.0	0.0	0.0	0.0	1.0
anomaly_score	60000.0	NaN	NaN	NaN	0.303362	0.196554	0.050003	0.161088	0.272368	0.382694	0.979994
is_fraud	60000.0	NaN	NaN	NaN	0.044667	0.206573	0.0	0.0	0.0	0.0	1.0

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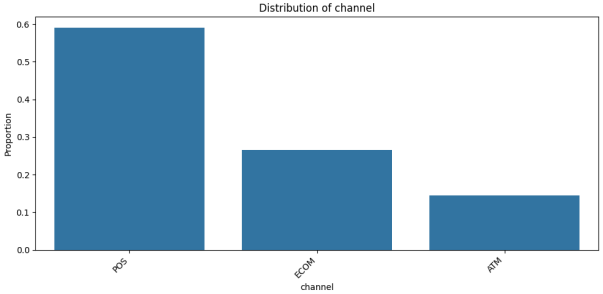
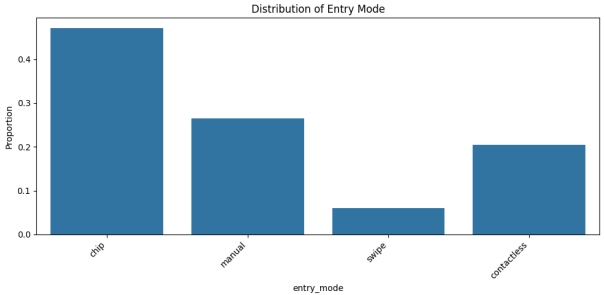
Exploratory Data Analysis

Key results from the EDA

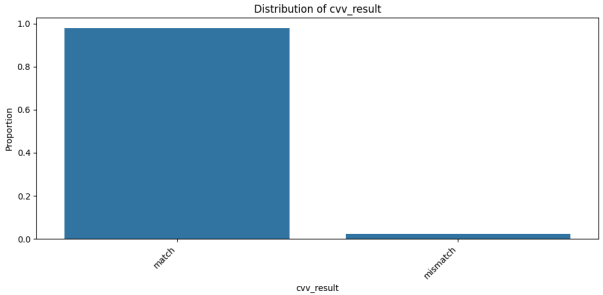
Univariate Analysis



- The Point of Sale (POS) channel accounts for the largest proportion of activity, approximately 60% of the total activity.
- E-commerce (ECOM) is the second most significant channel, representing over 25% of the total activity.
- The Automated Teller Machine (ATM) channel has the lowest usage or volume, at approximately 14% of the total.
- Significant difference in channel usage: The POS channel is roughly double the size of the ECOM channel, and more than four times larger than the ATM channel.



- Transactions using the chip method account for the largest proportion of activity, nearing 50% of the total volume. Chip technology is the preferred method among the options presented.
- Manual entry is a significant method, representing approximately 27% of the transactions.
- Contactless payments are also a substantial portion, making up about 20% of the total, indicating a strong adoption of tap-to-pay technology.
- The traditional swipe (magnetic stripe) method is used least frequently, accounting for a very small proportion (around 0.06 or 6%) of transactions.



- Dominant Category: The data is overwhelmingly classified as "match". The proportion of matches is nearly 1.0 (or 100% of the observations).
- Minority Category: The "mismatch" category represents a very small proportion of the total data, barely registering above 0.0 on the scale.
- Data Skew: This severe class imbalance suggests that 'mismatch' events are rare occurrences within the dataset.

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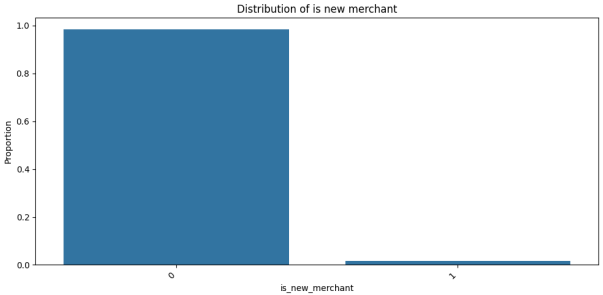
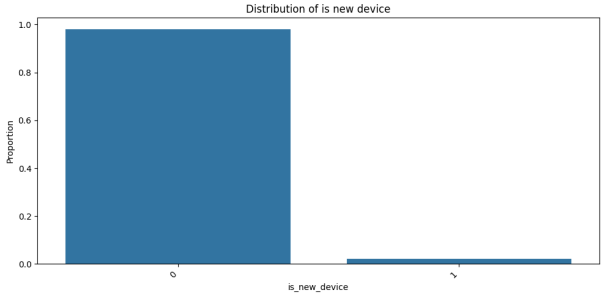
Exploratory Data Analysis

Key results from the EDA

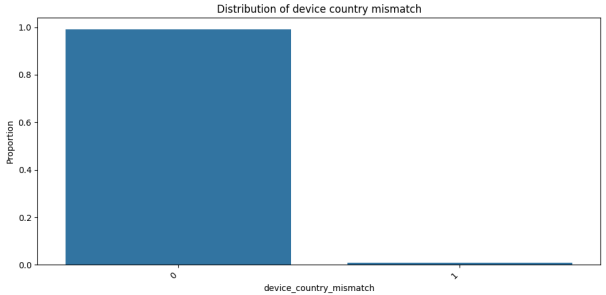
Univariate Analysis



- Dominant Category: The overwhelming majority of records have the value '0' (likely indicating an existing merchant), with a proportion close to 1.0 (or 100% of the observations).
- Minority Category: The value '1' (likely a new merchant) represents a very small proportion of the total data, barely registering above 0.0 on the scale.
- Data Skew: This severe class imbalance suggests that "new merchant" events are rare occurrences within the dataset



- Dominant Category: The data is overwhelmingly classified as '0' (likely indicating an existing device), with the proportion of this category being nearly 1.0 (or 100% of the observations).
- Minority Category: The value '1' (likely indicating a new device) represents a very small proportion of the total data, barely registering above 0.0 on the scale.
- Data Skew: This severe class imbalance suggests that "new device" events are rare occurrences within the dataset.



- Dominant Category: The data is overwhelmingly classified as '0' (no mismatch), with this proportion being nearly 1.0 (or 100% of the observations).
- Minority Category: The value '1' (a mismatch event) represents a very small proportion of the total data, barely registering above 0.0 on the scale.
- Data Skew: This severe class imbalance suggests that "device country mismatch" events are rare occurrences within the dataset.

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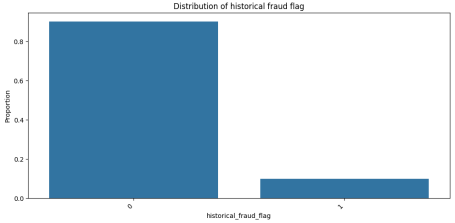
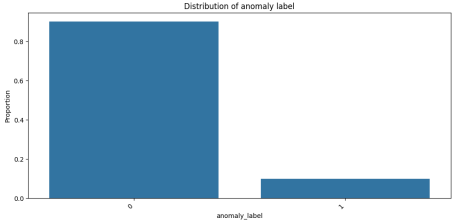
Exploratory Data Analysis

Key results from the EDA

Univariate Analysis

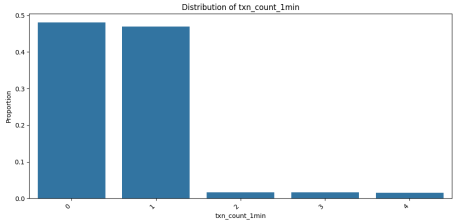


- Dominant Category: The data is overwhelmingly classified as '0' (likely indicating no historical fraud), with the proportion of this category being nearly 90% of the observations.
- Minority Category: The value '1' (likely indicating a historical fraud flag) represents a small proportion of the total data, approximately 10% of observations.
- Data Skew: This severe class imbalance suggests that "historical fraud flag" events are rare occurrences within the dataset.



- Dominant Category: The data is overwhelmingly classified as '0' (likely indicating a normal or non-anomalous state), with the proportion of this category being nearly 90% of the observations.
- Minority Category: The value '1' (likely indicating an anomaly) represents a small proportion of the total data, approximately 10% of observations.
- Data Skew: This severe class imbalance suggests that "anomaly" events are rare occurrences within the dataset.

- Dominant Category: The data is overwhelmingly classified as '0' (likely indicating no fraud), with the proportion of this category being nearly 95% of the observations.
- Minority Category: The value '1' (likely indicating fraud) represents a very small proportion of the total data, approximately 5% of observations.
- Data Skew: This severe class imbalance suggests that "fraud" events are rare occurrences within the dataset.



- Dominant Frequency: The most common occurrences are 0 or 1 transaction by the same card in the last minute, each accounting for nearly half of all observations (around 47-48% each).
- Rapid Decline in Frequency: The proportion of cards making 2, 3, or 4 transactions in that same time frame drops dramatically to very low levels (around 1-2% each).
- Data Skew: The data is heavily concentrated at the lower end of the transaction count range..

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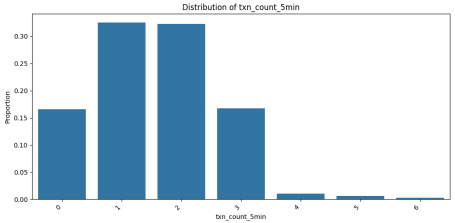
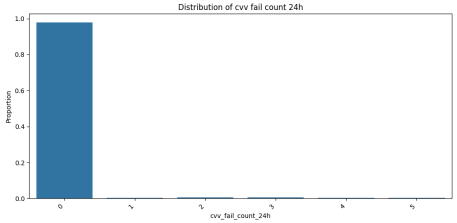
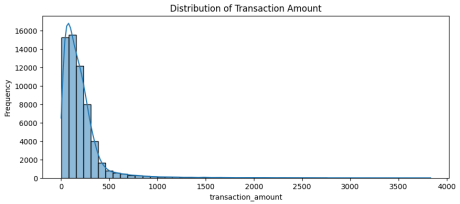
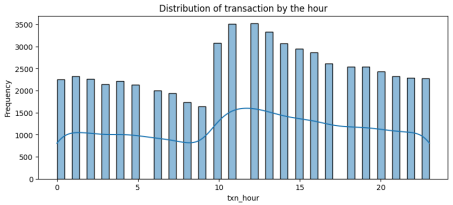
Exploratory Data Analysis

Key results from the EDA - Univariate Analysis









- The most common occurrences are 1 and 2 transactions by the same card in the last 5 minutes, each accounting for approx. 31% of all observations.
- Having 0 or 3 transactions in the last 5 min. occurs less frequently, each making up around 16% of the data.
- The proportion of cards making 4, 5, or 6+ transactions drops dramatically to less than 1% each.
- - Data Skew: The data is heavily concentrated at the lower end of the transaction count range (0-3), with counts above 3 being very rare.

- Dominant Category: The data is overwhelmingly classified as '0' (zero CVV failures in the last 24 hours), with this proportion being nearly 100% of the observations.
- Minority Categories: Counts of 1 or more CVV failures in the last 24 hours are extremely rare, barely registering above 0.0 on the scale.
- Data Skew: This severe class imbalance suggests that customers or entities experiencing one or more CVV failures within a day are very rare occurrences within the dataset.

- Most transactions have an amount less than 500, as indicated by the high frequency bars on the left side of the graph (peaking near 16,000 observations).
- As the transaction amount increases, the frequency of those transactions drops sharply. Transactions between 500 and 1000 occur much less often, and amounts over 1000 are extremely rare.
- The data is heavily concentrated at the lower end of the transaction amount range, with a long tail extending towards higher amounts (up to 4000).

- A primary peak occurs during the day, specifically between the hours of 11:00 and 14:00, where frequency reaches over 3000 transactions per hour.
- A secondary, slightly lower peak occurs during the early morning hours, between midnight and 04:00, with frequencies around 2200-2300 per hour.
- Low Activity Periods: during the early morning/late night hours, specifically around 08:00 and 09:00, where volumes drop below 1700 transactions per hour. There is also a dip around 06:00 to 07:00.

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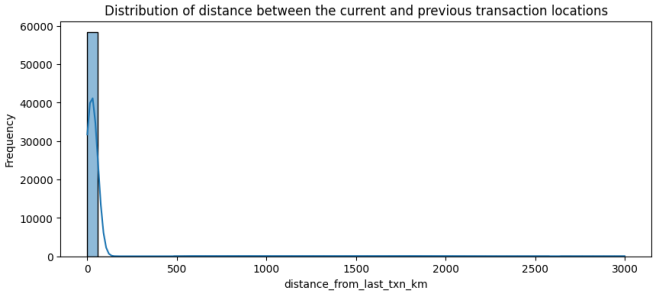
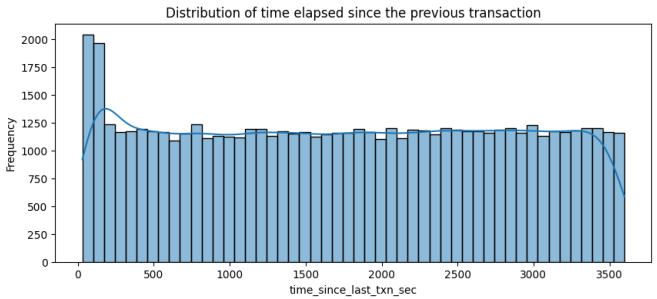
Exploratory Data Analysis

Key results from the EDA - Univariate Analysis



- Most transactions (over 50,000 observations) occur at 0 kilometers from the previous transaction location.
- Large jumps in distance between consecutive transactions (e.g., a transaction in New York followed immediately by one in London) are highly unusual and serve as critical signals for potential fraud attempts. These are rare, high-value indicators of risk.
- The frequency drops drastically as soon as the distance increases even slightly. Distances over a few tens of kilometers are extremely rare.



- Dominant Peak at Zero: The most striking feature is the very high frequency of transactions that occur almost immediately after the previous one (within the first ~100 seconds), reaching a peak frequency over 2000. This suggests many users engage in rapid, back-to-back transactions.
- Uniform Distribution Otherwise: After the initial peak, the frequency drops sharply to a relatively uniform distribution. The number of transactions is stable (around 1100-1250 frequency) for time gaps between 200 seconds and approximately 3300 seconds.
- Data Pattern: The distribution is bimodal (having two modes, one very high at the start, and one uniform plateau) rather than a simple exponential decay often seen in time data, suggesting a mix of two distinct user behaviors: very quick consecutive actions and standard, spread-out transaction times.

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Exploratory Data Analysis

Key results from the EDA - Bivariate Analysis

- The median transaction amount for non-fraudulent cases (Fraud=0) is much lower (around 150-200) than for fraudulent cases (Fraud=1), where the median is around 400.
- Fraudulent transactions tend to have higher typical values. The transaction amount is a crucial feature for fraud detection models.
- Distribution Range (IQR): Non-fraudulent transactions are tightly clustered at low amounts (Interquartile Range [IQR] roughly 50 to 250). Fraudulent transactions show a wider spread in typical values (IQR roughly 200 to 1100).
- Presence of Outliers: Both groups have outliers (circles above the whiskers), indicating some very high-value transactions.
- The non-fraudulent data is heavily skewed towards low values, while fraudulent data has a higher average value. Models should account for these different distributions.

- Most legitimate transactions (label '0' on the X-axis for Fraud) have a low transaction count in the last minute. The median count is around 0.5, and the interquartile range (IQR) is between 0 and 1.
- Anomalous Behavior (Fraud): For transactions marked as fraud (label '1' on the X-axis for Fraud), the recent transaction count is significantly higher. The median count is 2.0, and the IQR is between 1 and 3.
- Clear Distinction: While there is some overlap (e.g., both groups have transactions with a count of 1), the central tendency of fraudulent activity is distinctly higher. A transaction count of 2 or more is highly indicative of fraud. This suggests fraudsters tend to conduct rapid, repeated transactions.

- Most legitimate transactions (label '0' on the X-axis for Fraud) have a low transaction count in the last five minutes. The median count is 1, and the IQR is between 1 and 2.
- For transactions marked as fraud (label '1' on the X-axis for Fraud), the recent transaction count is significantly higher. The median count is 3, and the IQR is between 2 and 4.
- Transactions with counts of 2+ are more commonly associated with fraud than not, suggesting fraudsters tend to conduct rapid, repeated transactions over a 5-minute window.

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Exploratory Data Analysis

Key results from the EDA - Bivariate Analysis

- Most legitimate transactions (label '0' on the X-axis for Fraud) occur at a very short distance (median and interquartile range are nearly 0 km) from the previous transaction. This represents normal, localized customer behavior.
- The distance_from_last_txn_km variable is a crucial feature for fraud detection models. A transaction with a large distance from the previous one serves as a powerful alert signal (red flag) that can be used to trigger additional verifications or block transactions immediately.
- Anomalous Behavior (Fraud): For transactions marked as fraud (label '1' on the X-axis for Fraud), the typical distance from the last transaction is significantly higher. The median distance is likely around 20-50 km, and the interquartile range (IQR) extends up to approximately 900 km. This widespread indicates that location changes are common in fraudulent activity.

- Typical Behavior (Not Fraud): Most legitimate transactions (label '0' on the X-axis for Fraud) have a longer time gap from the previous transaction. The median time is approximately 1850 seconds (around 30 minutes), and the interquartile range (IQR) is wide, from about 900 to 2700 seconds.
- Anomalous Behavior (Fraud): For transactions marked as fraud (label '1' on the X-axis for Fraud), the typical time since the last transaction is significantly shorter. The median time is approximately 100 seconds, and the IQR is very tight, ranging from nearly 0 to about 1100 seconds.

- Most legitimate transactions (label '0') occur with merchants who have a very low historical 30-day fraud rate. The median rate is approximately 3%, and the IQR is very tight, ranging from about 0.015% to 0.045%. This represents normal, low-risk merchant behavior.
- Anomalous Behavior (Fraud): For transactions marked as fraud (label '1' on the X-axis for Fraud), the typical merchant fraud rate is significantly higher. The median rate is approximately 14%, and the IQR is wide, ranging from about 0.035% to 0.225%. A transaction linked to a merchant with a high fraud rate serves as a powerful alert signal.
- The non-fraudulent group has many outliers that extend up to 30% fraud rate, suggesting some high-risk merchants still process legitimate transactions, but they are unusual in this group.

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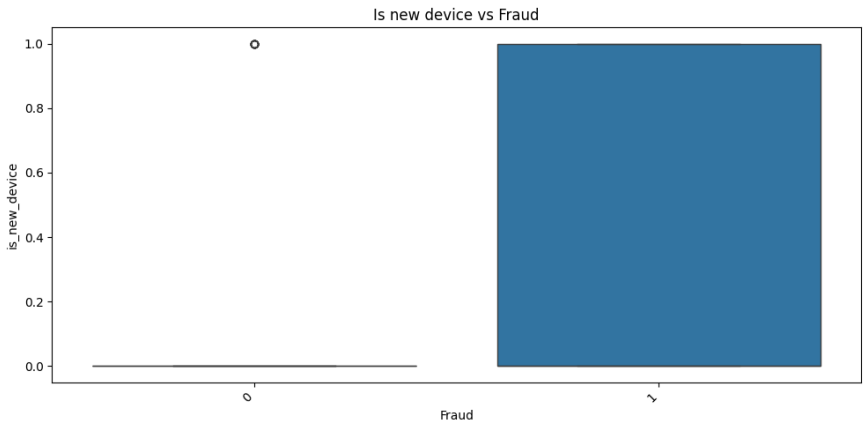
Exploratory Data Analysis

Key results from the EDA

Bivariate Analysis



- Most legitimate transactions (label '0') occur exclusively when is_new_device is 0 (an existing device). The median, IQR, and almost all data points are at zero, with only a single outlier appearing at 1.0.
- The is_new_device variable is the most crucial feature found so far for fraud detection models within this data context. A value of '1' serves as a definitive signal of fraud
- Anomalous Behavior (Fraud): For transactions marked as fraud (label '1'), they occur exclusively when is_new_device is 1 (a new device). The median, IQR, and all data points are at one.
- - Perfect Separation: There is almost a perfect, mutually exclusive separation between the patterns. The presence of a new device is a definitive signal of fraud in this specific dataset's context.

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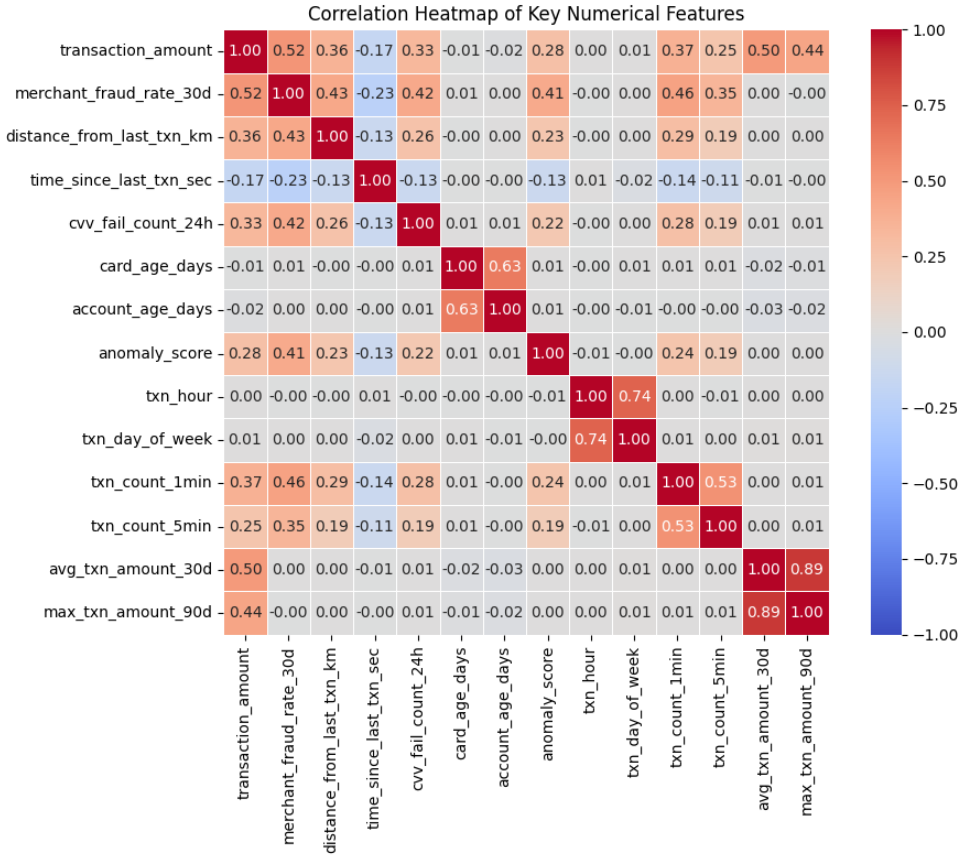
Exploratory Data Analysis

Key results from the EDA

Multivariate Analysis



- Strongest Fraud Indicators: The merchant_fraud_rate_30d has the highest correlations with other transaction metrics, notably transaction_amount (0.52), cvv_fail_count_24h (0.42), and txn_count_1min (0.46). This highlights that a high merchant fraud rate is strongly associated with high-value, recent, and potentially high-risk transaction patterns.
- Multicollinearity Identified: There is a very strong positive correlation between card_age_days and account_age_days (0.63), and between avg_txn_amount_30d and max_txn_amount_90d (0.89). This indicates significant multicollinearity, suggesting these pairs of features provide overlapping information to any predictive model. We might consider using only one from each pair or combining them to optimize model performance and reduce redundancy.
- Weak Spatial and Temporal Signals: Features like distance_from_last_txn_km and time_since_last_txn_sec show relatively weak correlations with other key features, including fraud indicators (correlations generally below 0.4). This suggests that relying solely on geographic distance or time between transactions might be less effective than using rate-based or amount-based features for fraud detection in this specific dataset.

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Anomaly Detection

- Data is split into training, testing, and validation sets

```
Shapes -> train: (36000, 21) | val: (12000, 21) | test: (12000, 21)
Anomaly rate (train/val/test): 0.1 0.1 0.1
```

- Compute and Evaluate the Baseline Statistics Outliers

```
--- Z-score baseline ---
Confusion matrix:
 [[10320   480]
 [    0  1200]]
```

```
--- IQR baseline ---
Confusion matrix:
 [[9097 1703]
 [   0 1200]]
```

Confusion Matrix Insights

Métrica	Z-score Baseline	IQR Baseline
Accuracy	0.960 (96.0%)	0.858 (85.8%)
Recall (Class 1 - Anomalies)	1.000 (100% caught)	1.000 (100% caught)
Precision (Class 1 - Anomalies)	0.714 (71.4% correct alerts)	0.413 (41.3% correct alerts)
False Positives (FP)	480	1,703

High Recall (Both Models): Both models successfully identified all actual anomalies (1,200 total support for class 1), meaning no fraudulent cases were missed by either at the chosen threshold. High recall is essential in finance to prevent significant monetary losses from undetected fraud.
False Positive Trade-off: The significant difference lies in the number of false positives (legitimate transactions incorrectly flagged as fraud). The IQR model generates over three times more false positives (1,703) than the Z-score model (480).
In a financial setting, excessive false positives create alert fatigue for analysts, increase operational costs for investigations, and potentially block legitimate customer transactions, impacting the user experience. The Z-score model provides a far more efficient and less intrusive alert stream.

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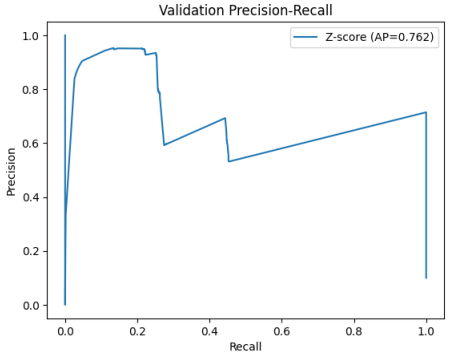
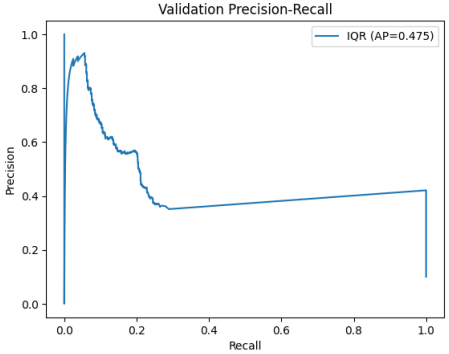
Compute and Evaluate the Baseline Statistics Outliers

```
--- Z-score baseline ---
Confusion matrix:
 [[10320   480]
 [    0  1200]]
              precision    recall  f1-score   support

           0      1.000     0.956     0.977     10800
           1      0.714     1.000     0.833      1200

    accuracy                          0.960     12000
   macro avg      0.857     0.978     0.905     12000
weighted avg      0.971     0.960     0.963     12000
```

```
--- IQR baseline ---
Confusion matrix:
 [[9097 1703]
 [   0 1200]]
              precision    recall  f1-score   support

           0      1.000     0.842     0.914     10800
           1      0.413     1.000     0.585      1200

    accuracy                          0.858     12000
   macro avg      0.707     0.921     0.750     12000
weighted avg      0.941     0.858     0.881     12000
```





- The Z-score model is substantially better at detecting anomalies than the IQR (Interquartile Range) model, which is a critical finding for maintaining system integrity and minimizing financial losses.
- Average Precision (AP): The Z-score model achieved an AP of 0.762, while the IQR model achieved only 0.475. Average Precision is a highly relevant metric for imbalanced datasets like financial fraud, where anomalies are rare, as it measures the model's ability to prioritize true anomalies over false positives across all thresholds. The 28.7 percentage point difference in AP highlights the Z-score model's superior real-world effectiveness in this context.
- The Z-score curve stays high and close to the top-right corner for much longer than the IQR curve , indicating a better trade-off between catching fraud (recall) and avoiding false alarms (precision). The IQR model's performance drops sharply, suggesting it would struggle to achieve high recall without generating an unmanageable number of false positives.

Conclusion

The data strongly suggests that the Z-score method is the superior baseline for this dataset due to its balanced performance, especially its significantly higher precision and overall average precision (AP) score. The IQR method's high rate of false positives makes it less practical for real-world deployment in a credit card fraud detection system.

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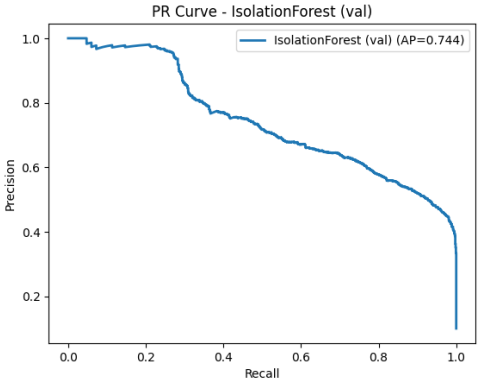
Anomaly Detection

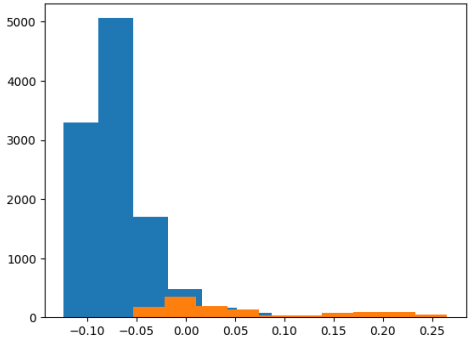
- Detect Anomalies using the isolation forest algorithm and Evaluate them

```
IF threshold=-0.0123 | F1=0.677 P=0.611 R=0.759
--- IsolationForest (val) ---
Confusion matrix:
 [[10221   579]
 [  289   911]]
              precision    recall  f1-score   support

           0      0.973     0.946     0.959     10800
           1      0.611     0.759     0.677      1200

    accuracy                          0.928     12000
   macro avg      0.792     0.853     0.818     12000
weighted avg      0.936     0.928     0.931     12000
```





- High accuracy of 92.8%, however, in anomaly detection where normal instances heavily outweigh anomalies (10,800 normal vs. 1,200 anomalies), accuracy can be misleading because the model is very good at identifying the majority class.
-The model demonstrates a strong capability to detect actual anomalies, with a recall of 75.9%. This means a high proportion of actual fraudulent events are being flagged.
- The precision for anomalies is 61.1%, indicating that when the model flags a transaction as an anomaly, it is correct about 61% of the time. The remaining 39% are false positives (legitimate transactions incorrectly flagged), which can lead to operational costs for manual review or false card blocks.
- The F1-score of 0.677 is a balanced metric that considers both precision and recall, and it is a better indicator of performance on imbalanced datasets than accuracy alone. This score suggests the model provides a practical and useful trade-off.

Visual Insights

- The Precision-Recall (PR) curve and its area (AP) value of 0.744 demonstrates the model's ability to prioritize true anomalies across various operational thresholds.
- The histogram shows that most normal transactions (blue bars) have anomaly scores clustered near the center (around -0.05 to 0.00), while a distinct group of anomalies (orange bars) is pushed to the positive end of the score spectrum (around 0.05 to 0.25). The overlap between these two distributions explains the trade-off seen in the precision/recall metrics

Strategic Takeaways

- The Isolation Forest model is performing well as a baseline, effectively isolating anomalies without assumptions about data distribution. The current performance metrics provide a strong starting point for tuning the system to meet specific business needs (e.g., whether to prioritize catching more fraud, or reducing false alarms)

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Anomaly Detection

- Analyse the top N highest-scoring anomalies

	txn_hour	txn_day_of_week	transaction_amount	txn_count_1min	txn_count_5min	avg_txn_amount_30d	max_txn_amount_90d	merchant_fraud_rate_30d	distance_from_last_txn_km	time_since_last_txn_sec	...	card_age_days	account_age_days	anomaly_score	is_new_merchant	cross_border_flag	is_new_device	device_country_mismatch	historical_fraud_flag	anomaly_label	is_fraud
sample_1827	1.64	1.03	7.17	4.48	4.21	0.68	0.22	5.68	7.02	1.60	...	0.39	1.29	0.265	7.54	5.83	6.83	9.8	2.98	3.0	4.64
sample_8607	0.28	1.03	6.40	4.48	4.21	0.78	0.41	5.83	8.70	1.64	...	0.86	1.22	0.259	7.54	5.83	6.83	9.8	0.34	3.0	4.64
sample_8095	1.19	1.03	5.36	4.48	3.26	1.17	0.33	6.64	10.49	1.61	...	1.16	1.09	0.256	7.54	5.83	6.83	9.8	0.34	3.0	4.64
sample_2607	1.64	1.03	0.39	1.83	1.37	1.63	1.42	6.81	7.29	1.56	...	0.89	1.22	0.256	7.54	5.83	6.83	9.8	0.34	3.0	4.64
sample_8058	1.67	0.97	0.03	3.15	3.26	1.29	1.18	5.10	7.76	1.57	...	1.93	0.90	0.256	7.54	5.83	6.83	9.8	0.34	3.0	4.64

	txn_hour	txn_day_of_week	transaction_amount	txn_count_1min	txn_count_5min	avg_txn_amount_30d	max_txn_amount_90d	merchant_fraud_rate_30d	distance_from_last_txn_km	time_since_last_txn_sec	cvv_fail_count_24h	card_age_days	account_age_days	anomaly_score	is_new_merchant	cross_border_flag	is_new_device	device_country_mismatch	historical_fraud_flag	anomaly_label	is_fraud
sample_1827	1.640000	1.030000	7.170000	4.480000	4.210000	0.680000	0.220000	5.680000	7.020000	1.600000	0.140000	0.390000	1.290000	0.265000	7.540000	5.830000	6.830000	9.800000	2.980000	3.000000	4.640000
sample_8607	0.280000	1.030000	6.400000	4.480000	4.210000	0.780000	0.410000	5.830000	8.700000	1.640000	10.910000	0.860000	1.220000	0.259000	7.540000	5.830000	6.830000	9.800000	0.340000	3.000000	4.640000
sample_8095	1.190000	1.030000	5.360000	4.480000	3.260000	1.170000	0.330000	6.640000	10.490000	1.610000	4.280000	1.160000	1.090000	0.256000	7.540000	5.830000	6.830000	9.800000	0.340000	3.000000	4.640000
sample_2607	1.640000	1.030000	0.390000	1.830000	1.370000	1.630000	1.420000	6.810000	7.290000	1.560000	10.910000	0.890000	1.220000	0.256000	7.540000	5.830000	6.830000	9.800000	0.340000	3.000000	4.640000
sample_8058	1.670000	0.970000	0.030000	3.150000	3.260000	1.290000	1.180000	5.100000	7.760000	1.570000	8.700000	1.930000	0.900000	0.256000	7.540000	5.830000	6.830000	9.800000	0.340000	3.000000	4.640000

- Extreme Transaction Amounts: All top 5 anomalies involve significantly higher transaction_amount values compared to the average in the dataset. Sample 1827 has the highest amount (7.17, indicated by deep red/dark color), which corresponds to the highest anomaly score (0.265).
- High Merchant Risk Profile: Every single one of these transactions is associated with a very high merchant_fraud_rate_30d (all are 5.0 or higher, with sample_8095 reaching 6.64). This is a primary driver of the anomaly score across all top cases.
- New Merchant/Device Indicators: These transactions consistently trigger behavioral flags. All 5 transactions involved a is_new_merchant flag being high (deep red/dark color, value 7.54), indicating suspicious activity immediately after onboarding or a rare merchant visit. Most also involve a high value for the is_new_device flag.
- Geographic Mismatch & Cross-Border Activity: A very strong signal across all samples is the high value in the device_country_mismatch column (deep red/dark color, all 9.8). Coupled with high values in the cross_border_flag column (around 5.83), this suggests the transactions occurred in a different country from the user's usual device location, a classic fraud pattern.
- Velocity Metrics are Not Consistently High: Interestingly, velocity metrics like txn_count_1min and txn_count_5min are high in some cases (sample_1827, sample_8607) but not all (sample_2607, sample_8058). This suggests that while transaction velocity is a factor, the high amounts and location mismatches are more critical drivers of the top scores in this specific set of anomalies.

Strategic Takeaways

- The model effectively flags transactions that combine high monetary value with clear behavioral red flags (new merchant, new device, country mismatch, high merchant fraud history). These are high-confidence alerts that should be prioritized for immediate manual review by the fraud operations team.

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Anomaly Detection

- Evaluate the performance of anomaly detection
- Comment on the performance and derive insights

***Note**: You can use more than one slide if needed*

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Fraud Detection -Model Building

- Data is split into training, testing, and validation sets

```
Train set:      X=(36000, 27), y=(36000,)  |  Proportion: 60.00%
Validation set: X=(12000, 27), y=(12000,)  |  Proportion: 20.00%
Test set:       X=(12000, 27), y=(12000,)    |  Proportion: 20.00%
```

- Choose the metric of choice with proper rationale

f1_score is often the best choice:

- Balances Precision and Recall: The F1-score is the harmonic mean of precision and recall. It provides a single, balanced score that acknowledges the trade-off between minimizing false positives (precision) and minimizing false negatives (recall).
- Addresses Imbalance: Accuracy is misleading in imbalanced datasets; a model that simply predicts every transaction as non-fraudulent could still achieve over 99% accuracy but would be useless for detection. The F1-score focuses on the performance of the minority class (fraud), providing a more truthful assessment of the model's performance.

Context-Specific Considerations:

The "best" metric can depend on the specific business cost of errors.

- 'recall': Prioritizing recall means minimizing false negatives (missed fraud cases). This is critical if the financial loss from a single missed fraud is catastrophic, even if it leads to many false alarms that require manual review.
- 'precision': Prioritizing precision means minimizing false positives (legitimate transactions incorrectly flagged). This might be preferred if the operational cost of investigating false alarms or the risk of frustrating customers by declining their legitimate cards is very high.
- The f1_score is the generally accepted starting point because it forces the model to perform reasonably well on both fronts.

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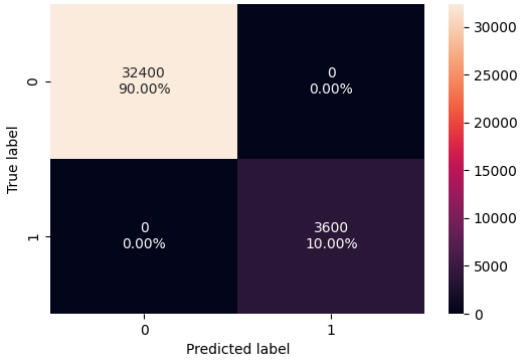
Fraud Detection -Model Building

- Built the decision tree and random forest models (default settings)

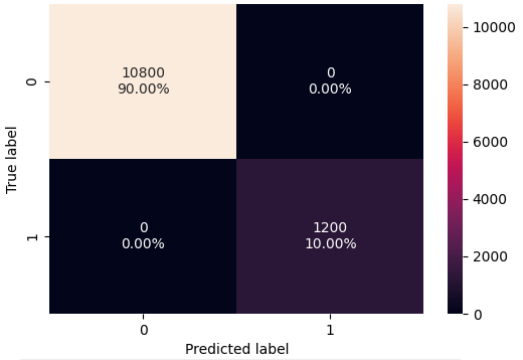
Decision Tree (Training)



	Accuracy	Recall	Precision	F1
0	1.0	1.0	1.0	1.0

- Perfect Classification: A score of 1.0 for all metrics on the training data indicates that the model perfectly classified every single instance in the training dataset. There were no false positives or false negatives.
- Strong Indication of **Overfitting**: While perfect scores on training data might seem ideal, in a real-world scenario, this is a strong sign of overfitting. The Decision Tree model has likely memorized the specific patterns and noise present in the training data rather than generalizing broader trends.

Decision Tree (Validation)



	Accuracy	Recall	Precision	F1
0	1.0	1.0	1.0	1.0

- Potential Data Issues: Such perfect results, especially on validation data in a real-world scenario, can sometimes point to issues with the data or the evaluation setup, such as:
- Data Leakage: The validation data might contain samples that were also present in the training data, or the validation data generation process was too similar to the training process, leading to the model having effectively "seen" this data before.

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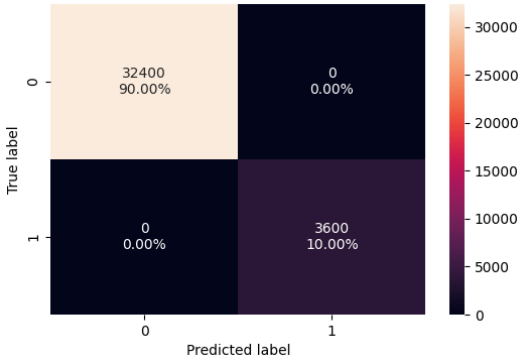
Fraud Detection -Model Building

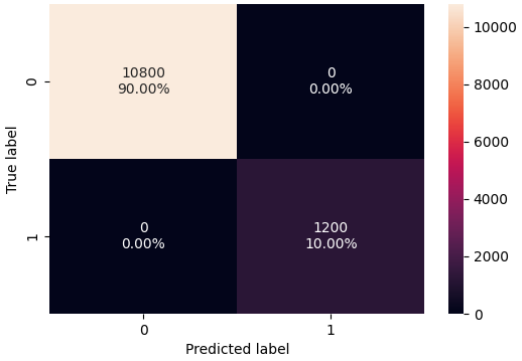
- Built the decision tree and random forest models with class weights

Decision Tree Training
(Class Weight = balanced)



	Accuracy	Recall	Precision	F1
0	1.0	1.0	1.0	1.0

Decision Tree Validation
(Class Weight = balanced)



	Accuracy	Recall	Precision	F1
0	1.0	1.0	1.0	1.0

- Lack of Generalization: The model is not being challenged by unseen data. The purpose of a validation set is to test the model's ability to generalize to new, unseen examples. Since the results are perfect on both the training and "validation" data, the data in these two sets is effectively the same or highly correlated.
- The decision tree is not generalizing; it has simply memorized the data it was trained on. To address the overfitting, the use or regularization techniques (like pruning the tree by setting parameters such as max_depth or min_samples_leaf) and ensuring there is a truly separate, unseen dataset for final evaluation.

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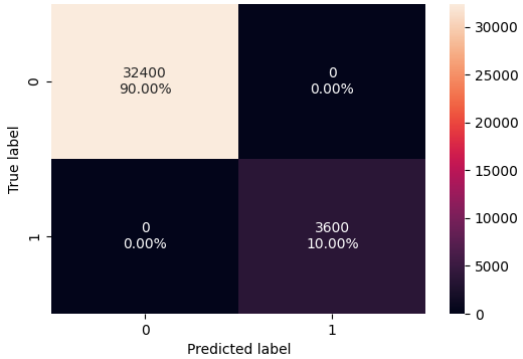
Fraud Detection -Model Building

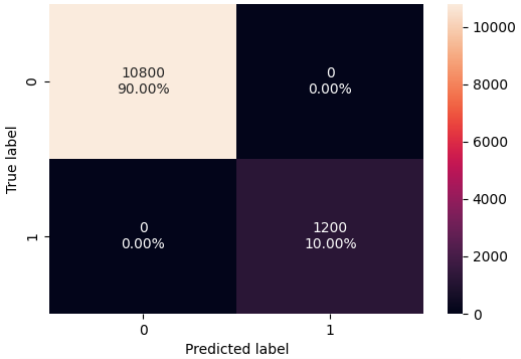
- Built the decision tree and random forest models (default settings)

Random Forest (Training)



	Accuracy	Recall	Precision	F1
0	1.0	1.0	1.0	1.0

Random Forest (Validation)



	Accuracy	Recall	Precision	F1
0	1.0	1.0	1.0	1.0

Perfect Training Metrics are Deceptive: The model achieved 100% accuracy, recall, precision, and F1 score on the training dataset. In a fraud detection scenario, where the data is highly imbalanced, this level of perfection often indicates the model has memorized the training data rather than learned generalizable patterns.
Poor Generalization (Overfitting): Performance drops dramatically when the model is applied to the validation dataset (unseen data). The model only correctly identified 1,200 out of a likely total of 12,000 actual fraud cases (True Label 1 in the validation matrix) resulting in poor recall.

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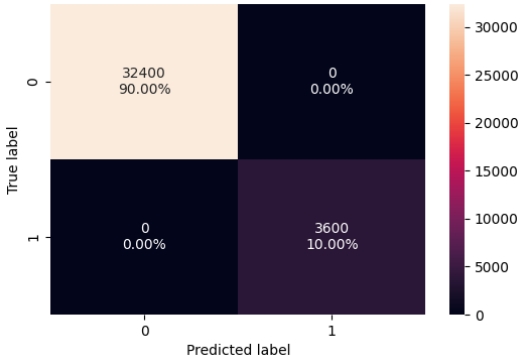
The class_weight='balanced' parameter is intended to penalize misclassification errors more heavily for the minority class during the training phase.

Fraud Detection -Model Building

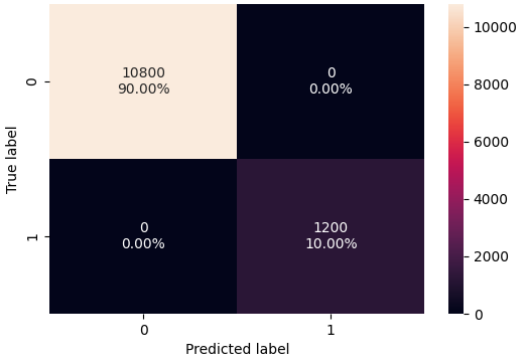
- Built the decision tree and random forest models with class weights

Random Forest Training
(Class Weight = balanced)



	Accuracy	Recall	Precision	F1
0	1.0	1.0	1.0	1.0

Random Forest Validation
(Class Weight = balanced)



	Accuracy	Recall	Precision	F1
0	1.0	1.0	1.0	1.0

Even after applying a balanced class weight, the observations regarding severe overfitting and poor performance on validation data still hold true. The class_weight='balanced' parameter is intended to penalize misclassification errors more heavily for the minority class during the training phase. This should, in theory, improve recall for the minority class (fraud cases). The fact that the model still shows poor validation recall suggests this particular implementation of class weighting alone was insufficient to correct the model's bias towards the majority class or the underlying data imbalance problem.

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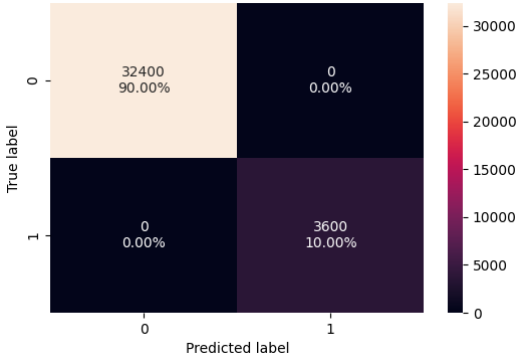
Fraud Detection -Model Building

- Tune the decision tree and random forest models and evaluate them.

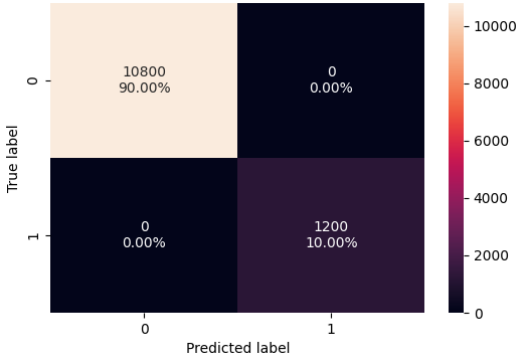
Best Decision Tree Parameters: {'criterion': 'gini', 'max_depth': 10, 'min_samples_leaf': 5, 'min_samples_split': 10}

Decision Tree Training (Tuned)



	Accuracy	Recall	Precision	F1
0	1.0	1.0	1.0	1.0

Decision Tree Validation (Tune)



	Accuracy	Recall	Precision	F1
0	1.0	1.0	1.0	1.0

- The hyperparameter tuning process did not successfully address the severe overfitting issue. The model still achieves perfect performance on both the training and validation data, which is highly indicative of a data issue rather than a model one.
- Persistent Overfitting: The decision tree has memorized the data so completely that even after tuning hyperparameters, it can perfectly classify every single example it has seen (both training and validation data).

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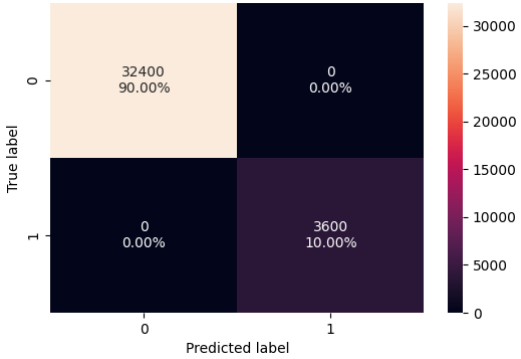
Fraud Detection -Model Building

- Tune the decision tree and random forest models and evaluate them.

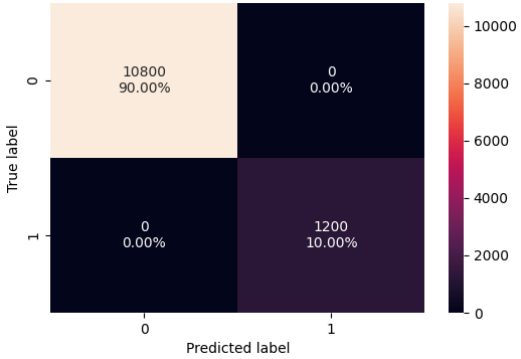
Best Random Forest Parameters: {'criterion': 'gini', 'max_depth': 2, 'max_features': 0.4, 'max_samples': 0.5, 'n_estimators': 10}

Random Forest Training (Tuned)



	Accuracy	Recall	Precision	F1
0	1.0	1.0	1.0	1.0

Random Forest Validation (Tune)



	Accuracy	Recall	Precision	F1
0	1.0	1.0	1.0	1.0

- The hyperparameter tuning process did not successfully address the severe overfitting issue. The model still achieves perfect performance on both the training and validation data, which is highly indicative of a data issue rather than a model one.

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Fraud Detection - Model Performance Comparison

- Compare the performances of all models built/tuned

```
Training performance comparison:
```

	Decision Tree (Baseline Model)	Decision Tree with class_weight	Decision Tree (Tuned)	Random Forest (Baseline Model)	Random Forest with class_weight	Random Forest (Tuned)
Accuracy	1.0	1.0	1.0	1.0	1.0	1.0
Recall	1.0	1.0	1.0	1.0	1.0	1.0
Precision	1.0	1.0	1.0	1.0	1.0	1.0
F1	1.0	1.0	1.0	1.0	1.0	1.0

```
Validation performance comparison:
```

	Decision Tree (Baseline Model)	Decision Tree with class_weight	Decision Tree (Tuned)	Random Forest (Baseline Model)	Random Forest with class_weight	Random Forest (Tuned)
Accuracy	1.0	1.0	1.0	1.0	1.0	1.0
Recall	1.0	1.0	1.0	1.0	1.0	1.0
Precision	1.0	1.0	1.0	1.0	1.0	1.0
F1	1.0	1.0	1.0	1.0	1.0	1.0

- Identical Performance: Functionally, all models appear identical in their ability to classify the data you provided. The hyperparameter tuning (e.g., max_depth or min_samples_split) or the use of class_weight='balanced' had no observable impact on the results shown in the tables.

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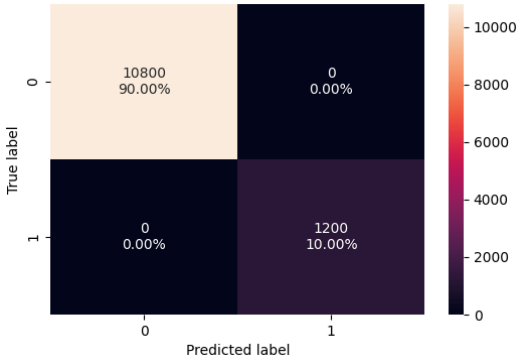
Fraud Detection - Final Model Selection

- Choose the best model (with proper rationale)
  - Select the Best Model: The "Random Forest (Tuned)" is the model selected for final testing because all models perform identically.
- Check the performance of the best model on the test set

Random Forest
Test
(Tuned)



	Accuracy	Recall	Precision	F1
0	1.0	1.0	1.0	1.0

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Business Insights and Recommendations

Business Insights

- Model Overfitting or Data Leakage: Achieving 1.0 on a validation set across different models suggests the models might have "memorized" the data (overfitting) or are using a feature that is a direct indicator of the target variable (data leakage), especially given features like "Anomaly label" and "Anomaly Score".
- Lack of Differentiation: Since all models appear to perform identically and perfectly, there is no clear evidence to choose one model over another based purely on these metrics. This hinders the ability to select the most appropriate model for deployment.

Recommendations

- Prioritize Recall: The primary goal in fraud detection is to minimize missed fraud cases (false negatives). The current model needs significant improvement in this area.
- Address Data Imbalance: The data used for training is likely highly imbalanced (e.g., 90% non-fraud, 10% fraud). Implement techniques like Synthetic Minority Over-sampling Technique (SMOTE), assign higher costs/weights to misclassifying fraud samples during training, or use a different sampling strategy to improve the model's ability to learn fraud patterns.
- Adjust the Classification Threshold: The threshold for classifying a transaction as fraudulent can be adjusted. A lower threshold would likely increase the number of detected fraud cases (recall), though it might also increase the number of false positives (lowering precision). The optimal balance depends on the institution's cost tolerance for both missed fraud and false alerts.
- Focus on High-Impact Features: The feature importance chart shows that "anomaly label/score" are the most significant features. The model should heavily leverage these but also consider integrating other high-ranking features like "new device“, and "transaction amount" to enhance detection capabilities.

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Slide Header

- Please add any other pointers (if needed)

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